

TESTING OF SOLAR-BASED DEVICES AND PANELS

With booming solar industry there is a need for measurement instruments to test the various components of solar-powered systems



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Solar energy has been harnessed by humans since ancient times using a wide range of ever-evolving technologies. The growing environmental concerns and demand for alternatives to fossil fuel-based energy sources have increased interest in solar cells as a long-term, exhaustless, environment-friendly and reliable energy technology. Continuous efforts to develop various types of solar cells are being made in order to produce solar cells with improved efficiencies at a lower cost, thereby taking advantage of the vast amounts of free energy available from the sun. Solar technology is being used in resi-

dential, commercial and military applications including space heating and cooling through solar architecture, potable water via distillation and disinfection, daylighting, solar hot water, solar cooking, and high-temperature process-heat for industrial purposes.

There has been an explosive growth in the solar industry, which in turn has intensified the need for measurement solutions for testing of various components of solar powered systems. Measurement solutions come in two main forms—complete turn-key solutions and test-system building blocks that must be fitted together and wrapped in software. They cater to the requirements during research and development, manufacturing

processes, deployment of the system on ground and finally maintenance of the system.

The major component in solar energy-based systems is a solar or photovoltaic (PV) cell. It is a solid-state device, basically a p-n junction with a very large light-sensitive area that converts the light energy directly into electricity by the photovoltaic effect. To increase their utility, dozens of individual PV cells are interconnected in a sealed, weather-proof package called a module. To achieve the desired voltage and current, modules are wired in series and parallel in a PV array, or what is called a solar panel. The flexibility of the modular PV system allows designers to create solar power systems that can meet a wide variety of electrical needs. PV cells can be modeled as a current source in parallel with a diode. When there is no light present to generate any current, the PV cell behaves like a diode. As the intensity of incident light increases, current is generated by the PV cell.

Most solar cell parameters can be obtained from simple I-V measurements. I-V testing systems include the light source, measurement electronics, computer and software needed to measure solar cell's I-V curves (refer figure on next page). The solar simulator illuminates the test device while the electronic load sweeps the cell voltage from a reverse-bias condition through the power quadrant, and beyond the open circuit voltage (V_{oc}). The system's computer gathers data, calculates solar cell parameters, generates printable test reports, and saves test data in text files. Solar cells